ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** independent variable	The quantity changed by the investigator is the independent variable.
2	**(a)** a question or curiosity about an observation	Questions drive investigations.
3	**(b)** length of the pendulum	Length is the quantity that is changed.
4	**(a)** dependent variable	It is the quantity that is measured as the result.
5	**(a)** a flexible approach that may be repeated and revised	Real investigations loop back and revise earlier steps.
6	**(c)** check the method and repeat the experiment	Differences must be investigated.
7	**(d)** thickness of the rolled dough	All factors other than the one being studied are kept constant.
8	**(d)** revise the hypothesis and test again	Honest reporting is essential.
9	**(b)** curiosity and willingness to question	Science asks for evidence.
10	**(c)** hypothesis	A hypothesis is a testable guess.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Testable explanation or guess before an experiment (1); e.g. a puri puffs more in hotter oil (1).	1+1
12	To check that the result is not a chance event (1); To get a more dependable (average) value (1).	1+1
13	Keep long hair and loose clothing tied back (1); Use tongs or holders; never leave the flame unattended (1).	1+1
14	A factor or quantity that can change (1); Independent (changed) and dependent (measured) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Question and hypothesis: hotter oil makes puffing faster (1); Vary temperature; keep dough, thickness and size same (1); Measure time with stopwatch, repeat, record in a table (1).	1+1+1
16	Real work often moves back and forth (1); A surprising result may send scientists back to change the question (1); So the steps are repeated and revised (1).	1+1+1
17	A test in which only one factor is changed (1); Others are kept the same (1); Otherwise we cannot tell which factor caused the effect (1).	1+1+1
	OR: Temperature of the oil (1); Amount and kind of flour and water in the dough (1); Size of the puri and time kept in oil (1).	1+1+1
18	Curiosity and asking questions (1); Honesty in recording and reporting data (1); Patience and willingness to repeat and revise (1).	1+1+1
	OR: Everyone gets comparable values (1); Results can be repeated and checked by others (1); Using body parts or guesswork gives different values (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Question stated clearly (1); Hypothesis and independent/dependent variables (1); Constants: oil temperature, thickness, size, frying time (1); Procedure: vary water, fry, observe puffing/time (1); Repeat and record in a table (1).	1+1+1+1+1
	OR: Parallax or wrong reading of scale (1); Faulty or uncalibrated instrument (1); Repeat the readings (1); Take the average (1); Use proper instrument and procedure (1).	1+1+1+1+1
20	Observe and ask a question (1); Form a hypothesis (1); Plan and perform a fair test (1); Analyse data and conclude (1); New questions arise, so steps repeat (1).	1+1+1+1+1
	OR: Tie hair and wear an apron (1); Use goggles/tongs (1); Do not leave the flame unattended (1); Keep water away from hot oil (1); Know the fire-safety steps and report accidents (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up the **independent** variable (changed by you) with the **dependent** variable (measured).
- Changing more than one factor at a time and still calling the test fair.
- Writing a hypothesis as a question or an opinion instead of a testable statement.
- Taking a single reading and skipping repeats; forgetting to average readings.
- Reading a scale or thermometer from the side (parallax) and omitting units in the answer.
- Drawing a conclusion that the data do not support, or generalising from one trial.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** organise data and see patterns	Tables show trends clearly.
2	**(b)** away from you and others	Hot liquid may spurt out.
3	**(a)** Does the amount of water in the dough change how well a puri puffs?	Only a question that can be answered by observation or measurement is scientific.
4	**(d)** show the relation between two quantities clearly	A graph reveals trends visually.
5	**(c)** eyes	They guard against splashes.
6	**(a)** qualitative observation	No number is involved.
7	**(d)** changing the data so that it matches the hypothesis	Fabricating data is dishonest.
8	**(a)** fair test	This is the idea of a controlled or fair test.
9	**(a)** quantitative observation	It is a number with a unit.
10	**(d)** a thermometer	Senses can mislead and can be unsafe.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Stopwatch measures time (1); Thermometer measures temperature of oil (1).	1+1
12	A statement answering the question of the investigation (1); It must be based on the data collected (1).	1+1
13	Then any change in the result can be linked to that factor (1); Changing many factors together makes the cause unclear (1).	1+1
14	Science depends on honest evidence (1); Wrong data leads to wrong conclusions for everyone who uses it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Observation is what we see or measure, e.g. the puri did not puff (1); Inference is the explanation drawn from it (1); e.g. the oil was probably not hot enough (1).	1+1+1
16	Puris puff faster as oil temperature rises (1); Independent variable: oil temperature (1); Dependent variable: time to puff (1).	1+1+1
17	Check the method for mistakes (1); Repeat the experiment (1); Revise the hypothesis and test again (1).	1+1+1
	OR: A single result may be affected by chance or error (1); Repeat the experiment several times (1); Take the average of the readings (1).	1+1+1
18	Others can check and repeat the experiment (1); Errors are corrected sooner (1); New ideas build on earlier work (1).	1+1+1
	OR: Crispness is directly sensed (1); 'Hot enough' is a judgement from other observations (1); Inferences can be wrong and need testing (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Qualitative: colour, smell, taste (1); Quantitative: time, temperature, mass (1); Examples with units (1); Quantitative data can be compared exactly (1); Qualitative data still helps in describing (1).	1+1+1+1+1
	OR: Data make the claim testable (1); Others can repeat and check (1); Without evidence beliefs may be wrong (1); e.g. false belief about a medicine harms health (1); Evidence-based thinking protects us (1).	1+1+1+1+1
20	Sum = 45 cm (1); Mean = 45 / 5 = 9 cm (1); Check the unusual value for error (1); Do not drop it without a reason (1); Repeat the observation if doubtful (1).	1+1+1+1+1
	OR: Hypothesis: hotter oil puffs faster (1); Variable: oil temperature (1); Observation: puffed in 6 s (1); Inference: oil was hot enough (1); Conclusion: puri puffs faster at high temperature (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** amount of water and type of soil	Only sunlight is varied.
2	**(a)** to make sure the result is reliable	Repetition reduces the chance that a single odd result misleads us.
3	**(b)** second	Time is measured in seconds.
4	**(c)** The puri in hotter oil will puff faster.	A prediction states what is expected before the test.
5	**(b)** stopwatch	A stopwatch measures time.
6	**(b)** measurements have some uncertainty	No measurement is perfectly exact.
7	**(b)** share, question and verify each other's work	Collaboration and checking build knowledge.
8	**(d)** conclusion	A conclusion is based on the data collected.
9	**(c)** get a more dependable value	Averaging reduces random errors.
10	**(c)** recorded and investigated further	Surprises often lead to discoveries.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Data are arranged neatly in rows and columns (1); Patterns and comparisons become easy to see (1).	1+1
12	Qualitative describes quality, e.g. the puri looks golden (1); Quantitative gives a number, e.g. it puffed in 8 s (1).	1+1
13	Steps are not always followed in one fixed order (1); Scientists may return to earlier steps, such as revising the question (1).	1+1
14	Answer depends on opinion (1); It cannot be measured or tested objectively (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum = 30 s (1); Average = 30 / 3 = 10 s (1); Average reduces the effect of small chance differences (1).	1+1+1
16	Do not let the bulb touch the vessel (1); Keep the eye level with the mercury or liquid column (1); Hold with a stand or holder, not by hand (1).	1+1+1
17	A hypothesis is proposed before the test (1); A conclusion is drawn after the data are analysed (1); A conclusion may confirm or reject the hypothesis (1).	1+1+1
	OR: Temperature is marked on one axis and time on the other (1); Points show a trend, e.g. time falls as temperature rises (1); The trend supports or rejects the hypothesis (1).	1+1+1
18	Cooking changes food by heat (chemical changes) (1); Pressure cookers cook faster by raising pressure and temperature (1); Refrigeration slows spoilage by lowering temperature (1).	1+1+1
	OR: Write observations in a table with units (1); Draw a graph or chart (1); Write a clear conclusion (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Independent variable: amount of water (1); Constants: type of seeds, temperature, light (1); Dependent variable: germination (1); Wet cotton gives water and air (1); Seeds in water get little air; dry cotton lacks water (1).	1+1+1+1+1
	OR: Time decreases as temperature increases (1); This supports the hypothesis (1); Few points give a weak pattern (1); Collect more data points (1); Test over a wider range (1).	1+1+1+1+1
20	Honesty: report even unexpected data (1); Curiosity: asks 'why' and starts investigations (1); Teamwork: groups share tasks and cross-check (1); Shared findings help others build on them (1); Science advances through verified work (1).	1+1+1+1+1
	OR: Observing fall of an apple or a rolling ball asks why (1); Noticing that boiling milk rises asks why (1); Questions lead to experiments (1); Experiments give laws and explanations (1); Curiosity keeps science alive (1).	1+1+1+1+1

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